

Network Traffic Analysis Using Wireshark

1. Introduction

Network Traffic Analysis involves capturing and examining network packets to understand communication between devices. Wireshark is a network protocol analyzer that helps inspect packet details, network endpoints, and communication sessions.

2. Objectives

- Capture and analyze network traffic.
- Identify protocols used in network communication.
- Examine source and destination endpoints.
- Analyze TCP and UDP conversations.
- Understand packet distribution and traffic volume.

3. Tools Used

- Wireshark
- PCAPNG network capture file

4. Methodology

1. Open the network capture file in Wireshark.
2. Examine the Protocol Hierarchy statistics.
3. Inspect IPv4 and IPv6 endpoints.
4. Analyze TCP and UDP conversations.
5. Review packet counts, byte counts, and communication ports.
6. Record observations and prepare the report.

5. Results and Observations

The analyzed capture contained 1,112 packets.

- IPv4: 138 packets
- IPv6: 963 packets
- TCP: 336 packets
- UDP: 582 packets
- TLS: 98 packets

- QUIC: 536 packets

The capture contained both IPv4 and IPv6 traffic. TCP and UDP were observed as transport protocols. TLS traffic was also identified, while QUIC traffic represented a significant part of the captured packets.

The endpoint analysis showed multiple network addresses communicating during the capture. Conversation statistics provided information about communication endpoints, port numbers, packet counts, and traffic volume.

6. Conclusion

Wireshark was used to examine a network traffic capture and identify its protocol distribution, endpoints, and communication sessions. The analysis provided practical experience in packet inspection and understanding common network protocols.

7. Evidence

Insert screenshots of:

1. Protocol Hierarchy Statistics

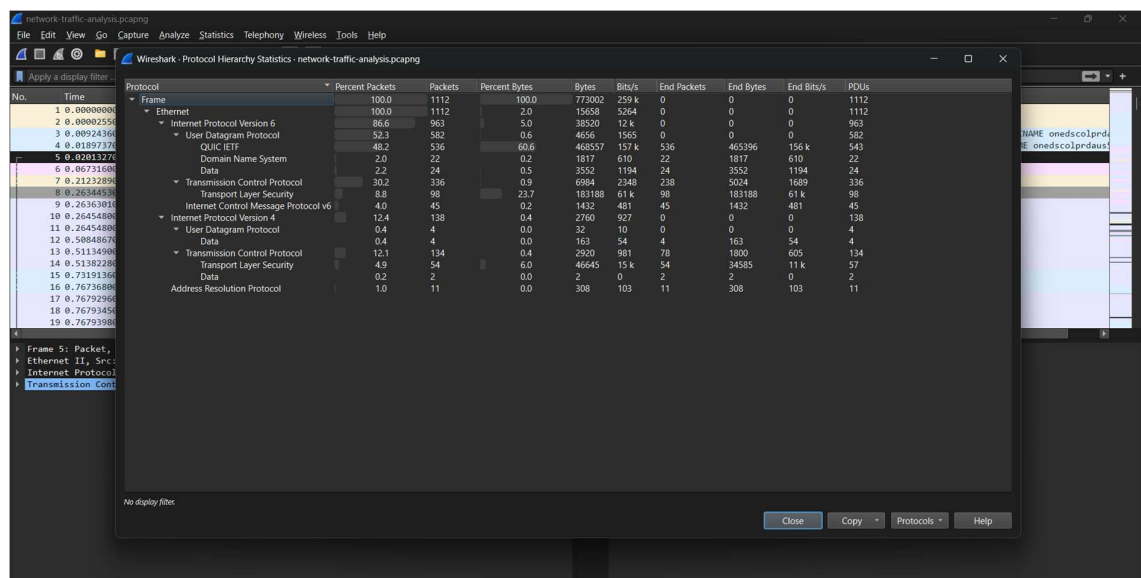
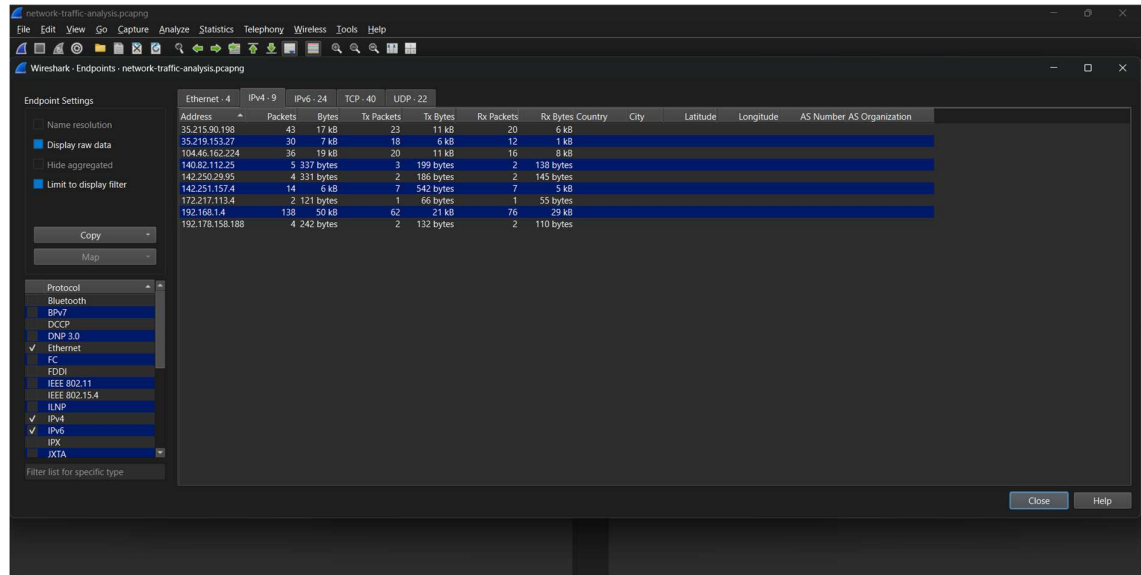


Figure 1: Protocol Hierarchy Statistics.

2. IPv4 Endpoints



Wireshark - Endpoints - network-traffic-analysis.pcapng

Endpoint Settings

- ☐ Name resolution
- ☒ Display raw data
- ☐ Hide aggregated
- ☒ Limit to display filter

Copy

Map

Protocol

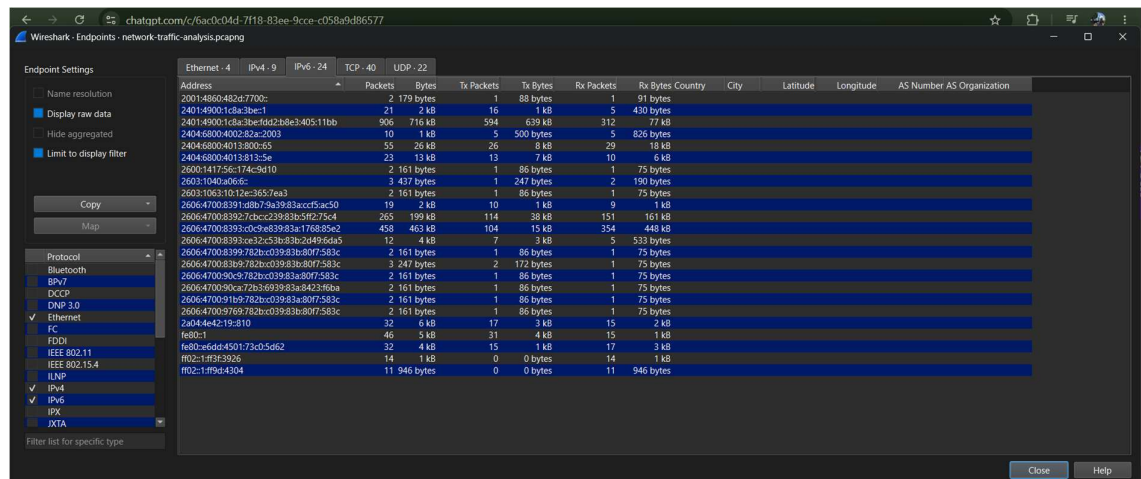
- Bluetooth
- IPv7
- DCCP
- DNP 3.0
- ✓ Ethernet
- FC
- FDI
- IEEE 802.11
- IEEE 802.15.4
- ILNP
- ✓ IPv4
- ✓ IPv6
- IPX
- JXTA

Filter list for specific type

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
35.215.90.198	43	17 kB	23	11 kB	20	6 kB						
35.219.153.27	30	7 kB	18	6 kB	12	1 kB						
104.46.162.224	36	19 kB	20	11 kB	16	8 kB						
140.82.112.25	5	337 bytes	3	199 bytes	2	138 bytes						
142.250.29.85	4	331 bytes	2	186 bytes	2	145 bytes						
142.251.157.4	14	6 kB	7	542 bytes	7	5 kB						
172.217.113.4	2	121 bytes	1	66 bytes	1	55 bytes						
192.168.1.4	138	50 kB	62	21 kB	76	29 kB						
192.178.158.188	4	242 bytes	2	132 bytes	2	110 bytes						

Figure 2: IPv4 Endpoints.

3. IPv6 Endpoints



Wireshark - Endpoints - network-traffic-analysis.pcapng

Endpoint Settings

- ☐ Name resolution
- ☒ Display raw data
- ☐ Hide aggregated
- ☒ Limit to display filter

Copy

Map

Protocol

- Bluetooth
- IPv7
- DCCP
- DNP 3.0
- ✓ Ethernet
- FC
- FDI
- IEEE 802.11
- IEEE 802.15.4
- ILNP
- ✓ IPv4
- ✓ IPv6
- IPX
- JXTA

Filter list for specific type

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
2001:4860:483d:7700::	2	179 bytes	1	88 bytes	1	91 bytes						
2401:4900:1c8a:3be:1	21	2 kB	16	1 kB	5	430 bytes						
2401:4900:1c8a:3be:1dd2:b8e3:405:11bb	906	716 kB	594	639 kB	312	77 kB						
2404:6800:4002:82a::2003	10	1 kB	5	500 bytes	5	826 bytes						
2404:6800:4013:800::65	55	26 kB	26	8 kB	29	18 kB						
2404:6800:4013:813::5e	23	13 kB	13	7 kB	10	6 kB						
2600:1417:56c:174c:9d10	2	161 bytes	1	86 bytes	1	75 bytes						
2603:1040:a06e::	3	437 bytes	1	247 bytes	2	190 bytes						
2603:1063:110:12a:365:7ea3	2	161 bytes	1	86 bytes	1	75 bytes						
2606:4700:8391:d8b7:9a39:83ac:cf5ac50	19	2 kB	10	1 kB	9	1 kB						
2606:4700:8392:7cbc:239:83b:5f2:75c4	265	199 kB	114	38 kB	151	161 kB						
2606:4700:8393:0c9e:839:83a:1768:85c2	459	463 kB	104	15 kB	354	448 kB						
2606:4700:8393:ae32:c3b:83b:c049:6da5	12	4 kB	7	3 kB	5	533 bytes						
2606:4700:8399:782b:c039:83b:807:583c	2	161 bytes	1	86 bytes	1	75 bytes						
2606:4700:83b9:782b:c039:83b:807:583c	3	247 bytes	2	172 bytes	1	75 bytes						
2606:4700:90c9:782b:c039:83b:807:583c	2	161 bytes	1	86 bytes	1	75 bytes						
2606:4700:90ca:72b3:69:83a:842:3f6aa	2	161 bytes	1	86 bytes	1	75 bytes						
2606:4700:91b9:782b:c039:83b:807:583c	2	161 bytes	1	86 bytes	1	75 bytes						
2606:4700:9769:782b:c039:83b:807:583c	2	161 bytes	1	86 bytes	1	75 bytes						
2604:4e42:19:810	32	6 kB	17	3 kB	15	2 kB						
fe80:c1	46	5 kB	31	4 kB	15	1 kB						
fe80:e6dd:4501:73cd:5d62	32	4 kB	15	1 kB	17	3 kB						
ff02::1:3f:3926	14	1 kB	0	0 bytes	14	1 kB						
ff02::1:93d:4304	11	946 bytes	0	0 bytes	11	946 bytes						

Figure 3: IPv6 Endpoints.

4. TCP Conversations

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
25.219.153.27	443	192.168.1.4	57393	30	7 kB	15	18	6 kB	12	1 kB	15.031272900
192.168.1.4	55487	35.215.90.198	443	29	9 kB	19	13	3 kB	16	6 kB	18.937402800
192.168.1.4	60351	35.215.90.198	443	14	8 kB	20	7	2 kB	7	5 kB	19.192209900
192.168.1.4	58252	104.46.162.224	443	36	19 kB	0	16	8 kB	20	11 kB	0.020132700
192.168.1.4	50325	140.82.112.25	443	5	337 bytes	2	2	138 bytes	3	199 bytes	5.180995700
192.168.1.4	57103	142.251.157.4	443	14	6 kB	4	7	5 kB	7	542 bytes	5.741028300
192.168.1.4	51517	172.217.113.4	443	2	121 bytes	3	1	55 bytes	1	66 bytes	5.302222700
192.168.1.4	60541	192.178.158.188	5228	2	121 bytes	7	1	55 bytes	1	66 bytes	13.135949800
192.168.1.4	63031	192.178.158.188	5228	2	121 bytes	8	1	55 bytes	1	66 bytes	13.185005500
2401:4900:1c8a3befdd2b8e3:405:11bb	61751	2600:1417:56c174c9d10	80	2	161 bytes	16	1	75 bytes	1	86 bytes	15.057769800
2401:4900:1c8a3befdd2b8e3:405:11bb	60942	2603:10a0:9d6de	443	3	437 bytes	6	2	100 bytes	1	247 bytes	12.434571600
2401:4900:1c8a3befdd2b8e3:405:11bb	57592	2603:1063:1012e:365:7ea3	443	2	161 bytes	11	1	75 bytes	1	86 bytes	14.354973900
2401:4900:1c8a3befdd2b8e3:405:11bb	65205	2606:4700:839f:08b7:9a39:83accd5ac50	443	19	2 kB	5	9	1 kB	10	1 kB	9.122783000
2401:4900:1c8a3befdd2b8e3:405:11bb	57212	2606:4700:839f:7dbcc039:83b:5f02:75c4	443	265	199 kB	18	151	161 kB	114	30 kB	18.788515100
2401:4900:1c8a3befdd2b8e3:405:11bb	64499	2606:4700:839f:7dbcc039:83b:5f02:75c4	443	2	161 bytes	14	1	75 bytes	1	86 bytes	14.033113700
2401:4900:1c8a3befdd2b8e3:405:11bb	58032	2606:4700:839f:782bxc039:83b:80f7:583c	443	3	247 bytes	17	1	75 bytes	2	172 bytes	16.611115700
2401:4900:1c8a3befdd2b8e3:405:11bb	63338	2606:4700:90c9:782bxc039:83b:80f7:583c	443	2	161 bytes	10	1	75 bytes	1	86 bytes	14.338823400
2401:4900:1c8a3befdd2b8e3:405:11bb	57881	2606:4700:90c9:782bxc039:83b:80f7:583c	443	2	161 bytes	9	1	75 bytes	1	86 bytes	14.276546700
2401:4900:1c8a3befdd2b8e3:405:11bb	64045	2606:4700:9109:782bxc039:83b:80f7:583c	443	2	161 bytes	12	1	75 bytes	1	86 bytes	14.482461300
2401:4900:1c8a3befdd2b8e3:405:11bb	51449	2606:4700:9769:782bxc039:83b:80f7:583c	443	2	161 bytes	13	1	75 bytes	1	86 bytes	14.607762100
2401:4900:1c8a3befdd2b8e3:405:11bb	55543	2a04:4e42:19:810	443	32	6 kB	1	15	2 kB	17	3 kB	4.665083500

Figure 4: TCP Conversations.

5. UDP Conversations

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
192.168.1.4	61297	142.250.29.95	443	4	331 bytes	5	2	145 bytes	2	186 bytes	4.692678600
2401:4900:1c8a3befdd2b8e3:405:11bb	61475	2001:4860:482d:7700:	443	2	179 bytes	4	1	91 bytes	1	88 bytes	3.448292400
2401:4900:1c8a3befdd2b8e3:405:11bb	56953	2404:6800:4002:82a:2003	443	10	1 kB	7	5	826 bytes	5	500 bytes	18.832792200
2401:4900:1c8a3befdd2b8e3:405:11bb	57425	2404:6800:4013:800c:45	443	55	26 kB	3	29	18 kB	26	8 kB	1.094072700
2401:4900:1c8a3befdd2b8e3:405:11bb	58033	2404:6800:4013:813:5e	443	23	13 kB	13	10	6 kB	13	7 kB	22.87557000
2401:4900:1c8a3befdd2b8e3:405:11bb	63755	2606:4700:839f:c0c9:e839:83a:1768:85e2	443	458	463 kB	9	354	448 kB	104	15 kB	19.140501900
2606:4700:839f:c0c9:e839:83a:1768:85e2	443	2401:4900:1c8a3befdd2b8e3:405:11bb	63727	12	4 kB	2	7	3 kB	5	533 bytes	0.719116600
fe80:c1	53	fe80:e6dd:4501:73cd:5d62	49665	1	297 bytes	0	1	297 bytes	0	0 bytes	0.000241600
fe80:c1	53	fe80:e6dd:4501:73cd:5d62	63344	1	242 bytes	1	1	242 bytes	0	0 bytes	0.018973700
fe80:e6dd:4501:73cd:5d62	50907	fe80:c1	53	2	340 bytes	11	1	100 bytes	1	240 bytes	20.187301700
fe80:e6dd:4501:73cd:5d62	58205	fe80:c1	53	8	902 bytes	6	4	405 bytes	4	497 bytes	18.757571700
fe80:e6dd:4501:73cd:5d62	58476	fe80:c1	53	6	774 bytes	8	3	316 bytes	3	458 bytes	18.912423500
fe80:e6dd:4501:73cd:5d62	60225	fe80:c1	53	2	298 bytes	10	1	100 bytes	1	198 bytes	20.186751200
fe80:e6dd:4501:73cd:5d62	64541	fe80:c1	53	2	328 bytes	12	1	100 bytes	1	228 bytes	20.187743200

Figure 5: UDP Conversations

8. Limitations

The results describe only the supplied packet capture. They do not establish that the network contains malicious activity or represent all traffic on the network.